Learning from examples and checking mistakes

Teacher reference

Original classroom activities for Year 2. Use teacher-led demonstrations, prepared predictions and non-personal pictures.

Lesson 1 Examples and labels

An AI classifier can use labelled examples to learn patterns. This classroom card activity is a model of the idea; pupils sorting cards are not a real AI system.

Prepare several fish pictures and several non-fish pictures. Include different shapes, colours and sizes. Label each example and ask what features help tell them apart.

Show a new picture and ask pupils to predict a label. Compare using varied examples with repeatedly showing the same picture.

Lesson 2 A prediction can be wrong

Prepare a results sheet with some correct and incorrect labels. Include a fish-shaped toy labelled real fish.

Pupils mark which predictions need checking. Compare the picture, its correct label and the system's guess. Explain why a person should check unfamiliar examples.

Do not describe a confident answer as proof. Ask what extra examples or information might help.

Check

Can the pupil explain why varied examples help and identify one incorrect prediction? Accept an oral explanation or drawing.

Teacher sources

AI4K12 grade-band progression charts (K-2): https://ai4k12.org/gradeband-progression-charts/

UNICEF Guidance on AI and Children: https://www.unicef.org/innocenti/reports/policy-guidance-ai-children